TCS NQT 2027 · SECTION 1 OF 5 · FOUNDATION

Numerical Ability Complete Guide

Every topic explained from zero: rules, formulas, diagrams, solved examples and practice questions.

14	45+	140+	20
topics from basic to advanced	solved examples	practice questions	question mini test

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About this guide

Numerical Ability is the first section of the TCS NQT Foundation test. It has 20 questions in 25 minutes (about 75 seconds each), moderate difficulty, no negative marking, and some questions are TITA (you type the exact answer).

How each topic is organised

- **Start from zero:** the idea in plain words.
- **Formulas:** a yellow box to revise quickly.
- **Diagrams:** pictures, charts and graphs.
- **Solved examples:** step by step.
- **Shortcuts and mistakes** to avoid.
- **Practice questions:** Basic, Medium and Advanced. Answers are at the end.

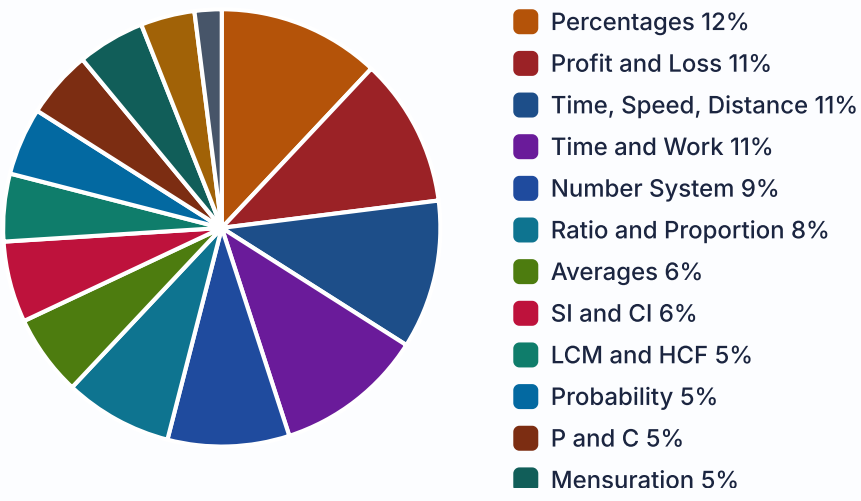
How to study

- Read one topic, then solve its practice set without looking at the answers.
- Check answers, and note wrong ones in an error log.
- Revise the formula boxes daily.
- After all topics, take the 20-question mini test in 25 minutes.
- Follow the priority: do **Must-do** topics first.

Topic priority and contents

1. Number System	Must-do	2. LCM and HCF	Must-do
3. Percentages	Must-do	4. Profit and Loss	Must-do
5. Time, Speed and Distance	Must-do	6. Time and Work	Must-do
7. Ratio and Proportion	Must-do	8. Averages	Important
9. Simple and Compound Interest	Important	10. Probability	Important
11. Permutation and Combination	Important	12. Mensuration	Important
13. Mixtures and Alligation	Important	14. Ages, Simplification and Approximation	Good to know

Where to spend your practice time



Suggested share of practice time per topic. This is our preparation guidance, not an official TCS weightage.

Section strategy: 20 questions in 25 minutes

Time plan

Phase	Time	What to do
Pass 1	14 min	Solve easy and medium questions first
Pass 2	8 min	Return to skipped questions
Final	3 min	Guess any left blank (no negative marking)

Rules of this section

- Question switching is allowed inside the section.
- You cannot return after the section timer ends.
- No negative marking: never leave a question blank.
- TITA: type the exact value (check units and rounding).
- Use rough paper for working.

Suggested order of attempt

Percentages | **Profit and Loss** | **Averages** | **TSD, Work** | **Ratio, SI/CI** | **P and C, Prob.**

Green: quick wins · Amber: medium · Red: leave for the second pass

Guidance on what to attempt first. Always pick the questions you find easiest.

Speed habits

- Memorise squares to 30, cubes to 15 and tables to 20.
- Use fractions for percentages (see the cheat sheet).
- Skip anything that takes over 2 minutes.
- Estimate first, then calculate only if options are close.

Where marks are lost

- Wrong base in percentages.
- Unit mistakes in speed and mensuration.
- Spending too long on one hard problem.
- Arithmetic slips under time pressure.

01 NUMERICAL ABILITY · MUST-DO
Number System

Every other topic in Numerical Ability sits on top of numbers. Here we learn what kinds of numbers exist, how to test divisibility quickly, how to find remainders and unit digits, and how to count factors. These ideas appear directly in questions and silently inside others (LCM, ratios, P and C).

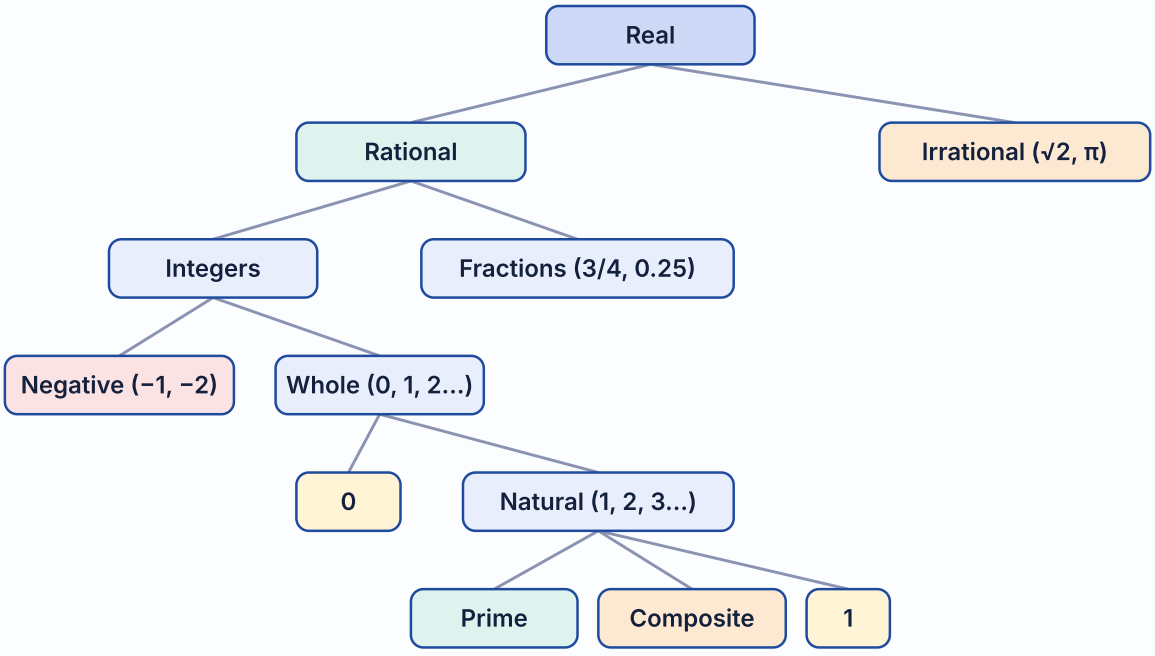
Start from zero

Types of numbers

Natural numbers are 1, 2, 3 ... **Whole** numbers add 0. **Integers** add negatives (... −2, −1, 0, 1, 2 ...). **Rational** numbers can be written as p/q (like 3/4, 0.25, 5). **Irrational** numbers cannot (√2, π). A **prime** has exactly two factors (1 and itself): 2, 3, 5, 7, 11 ... 2 is the only even prime. A **composite** has more than two factors. 1 is neither prime nor composite.

Divisibility rules

By	Rule	Example
2	Last digit even	4,718 ✔
3	Digit sum divisible by 3	2,451: 2+4+5+1=12 ✔
4	Last two digits divisible by 4	3,516: 16 ✔
5	Last digit 0 or 5	1,245 ✔
6	Divisible by 2 and 3	732 ✔
8	Last three digits divisible by 8	5,120: 120 ✔
9	Digit sum divisible by 9	4,383: 18 ✔
11	(Sum of odd-place digits) − (sum of even-place digits) is 0 or a multiple of 11	918,291 ✔



Family tree of numbers

Remainders

When a is divided by n, the leftover is the **remainder**. A key idea: the remainder of a product equals the remainder of the product of remainders. Example: 17 × 23 divided by 5. 17 leaves 2, 23 leaves 3, so 2 × 3 = 6, which leaves 1. For powers, find the repeating pattern: $2^1=2$, $2^2=4$, $2^3=8\equiv1$ (mod 7), so powers of 2 repeat every 3 steps when divided by 7.

Unit digit and cyclicity

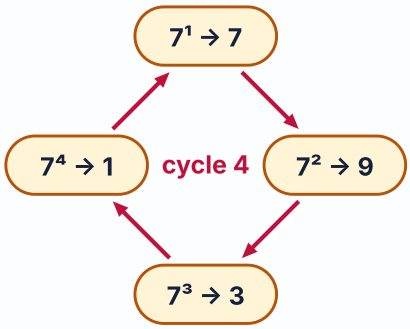
Powers of a number repeat their last digit in a cycle. Digits 0, 1, 5, 6 never change. Digits 4 and 9 cycle in 2 steps. Digits 2, 3, 7, 8 cycle in 4 steps. Divide the power by the cycle length; the remainder tells the position (remainder 0 means the last item of the cycle).

Factors

Write N as a product of prime powers: $N = p^a \times q^b \times r^c$. Then the number of factors is $(a+1)(b+1)(c+1)$. The sum of factors is $(p^0+\ldots+p^a)(q^0+\ldots+q^b)(r^0+\ldots+r^c)$. Numbers like 36, 49, 100 (perfect squares) have an odd number of factors.

Trailing zeros and bases

Each trailing zero needs a pair 2 × 5, and 5s are rarer, so the zeros in n! equal $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$. To convert decimal to binary, divide repeatedly by 2 and read the remainders from bottom to top.



Unit digit of 7^n repeats: 7, 9, 3, 1

7^{123}: 123 ÷ 4 leaves remainder 3

3rd item of the cycle → unit digit = 3

Cycle lengths: 0,1,5,6 → fixed · 4,9 → 2 · 2,3,7,8 → 4

Unit-digit cycle of 7 (cyclicity)

Formula box

- Number of factors of $N = p^a q^b r^c \to (a+1)(b+1)(c+1)$
- Sum of factors = $[(p^{a+1}-1)/(p-1)] \times [(q^{b+1}-1)/(q-1)] \times \ldots$
- Trailing zeros of n! = $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$
- Dividend = Divisor × Quotient + Remainder
- Sum of first n natural numbers = $n(n+1)/2$
- Sum of first n odd numbers = n^2; first n even numbers = $n(n+1)$
- $(a+b)^2 = a^2+2ab+b^2$, $(a-b)^2 = a^2-2ab+b^2$, $a^2-b^2 = (a+b)(a-b)$

Step-by-step method

1. Check divisibility with the rules before dividing.
2. For remainders of products or sums, reduce each number first, then combine.
3. For unit digits of a^n, take the unit digit of a, find its cycle, reduce n modulo the cycle length.
4. For factor counts, always write the prime factorisation first.

Solved examples

Example 1. Find the unit digit of 7^{123}.

Solution: The unit digit of powers of 7 cycles 7, 9, 3, 1 (length 4). Divide 123 by 4: remainder 3. The 3rd item of the cycle is 3.

Answer: 3

Example 2. How many factors does 360 have? What is their sum?

Solution: $360 = 2^3 \times 3^2 \times 5^1$. Count = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = 24$. Sum = $(2^4-1)/(2-1) \times (3^3-1)/(3-1) \times (5^2-1)/(5-1) = 15 \times 13 \times 6 = 1170$.

Answer: 24 factors, sum 1170

Example 3. How many trailing zeros are in 100!?

Solution: $\lfloor 100/5 \rfloor = 20$, $\lfloor 100/25 \rfloor = 4$, $\lfloor 100/125 \rfloor = 0$. Total = 20 + 4 = 24.

Answer: 24

Shortcuts

- Memorise primes up to 100 (25 of them).
- For remainder by 9, use the digit sum. For remainder by 11, use the alternating sum.
- A number is divisible by 12 if it is divisible by both 3 and 4 (not by 2 and 6).

Common mistakes

- Treating 1 as a prime number.
- Using cycle length 4 for digit 4 or 9 (their cycle is 2).
- Forgetting that remainder 0 in cyclicity means the last item of the cycle.

Practice questions: Number System

1. Find the unit digit of 3^{57}. Basic
2. What is the remainder when 2^{10} is divided by 7? Basic
3. Is 918,291 divisible by 11? (answer Yes or No) Basic
4. How many factors does 180 have? Basic
5. How many trailing zeros are in 50!? Medium
6. Convert 45 (decimal) to binary. Medium
7. Find the smallest 4-digit number divisible by 12. Medium
8. What is the remainder when 17 × 23 is divided by 5? Medium
9. Find the last two digits of 7^{100}. Advanced
10. Find the sum of all factors of 28. Advanced

Answers: see the answer key at the end of this guide.

02 NUMERICAL ABILITY · MUST-DO
LCM and HCF

HCF (highest common factor) is the largest number that divides all the given numbers. LCM (lowest common multiple) is the smallest number that all of them divide. Real situations: cutting ribbons into equal pieces (HCF) and bells ringing together (LCM).

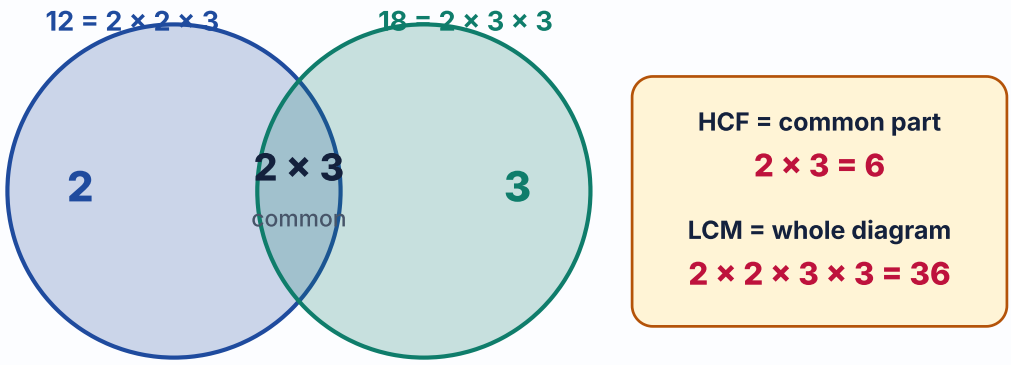
Start from zero

Prime factorisation method

Write each number as a product of primes. HCF takes the **lowest power** of each common prime. LCM takes the **highest power** of every prime that appears. Example: 12 = $2^2 \times 3$ and 18 = 2×3^2. HCF = $2^1 \times 3^1$ = 6. LCM = $2^2 \times 3^2$ = 36.

Product rule

For two numbers: **a × b = HCF × LCM**. So if you know three of these four values you can find the fourth. This rule is for two numbers only.



Prime factors of 12 and 18: HCF is the overlap, LCM is everything

HCF and LCM of fractions

HCF of fractions = HCF of numerators / LCM of denominators. LCM of fractions = LCM of numerators / HCF of denominators (fractions in lowest terms).

Word problem patterns

Greatest number dividing a, b, c leaving the same remainder r → HCF of (a−r, b−r, c−r). **Leaving remainders r_1, r_2, r_3** → HCF of ($a-r_1$, $b-r_2$, $c-r_3$). **Smallest number leaving remainder r with divisors x, y, z** → LCM(x, y, z) + r. **Events repeating together** (bells, lights, runners on a track) → LCM of their time gaps. **Equal division** (tiles, ribbons, rows) → HCF.

Formula box

- HCF × LCM = product of the two numbers
- HCF ≤ every given number ≤ LCM
- HCF of fractions = HCF(numerators) / LCM(denominators)
- LCM of fractions = LCM(numerators) / HCF(denominators)
- Smallest number leaving remainder r = LCM + r
- Greatest number leaving remainder r = HCF(a−r, b−r, ...)

Step-by-step method

1. Decide whether the story asks for the biggest equal piece (HCF) or the first common time (LCM).
2. Factorise each number into primes.
3. Apply lowest powers (HCF) or highest powers (LCM).
4. For remainder problems, adjust the numbers first, then use HCF or LCM.

Solved examples

Example 1. Find the HCF and LCM of 12 and 18.

Solution: $12 = 2^2 \times 3$, $18 = 2 \times 3^2$. HCF $= 2 \times 3 = 6$. LCM $= 2^2 \times 3^2 = 36$. Check: $6 \times 36 = 216 = 12 \times 18$.

Answer: HCF 6, LCM 36

Example 2. Find the greatest number that divides 60 and 76 leaving remainder 4 each time.

Solution: Subtract the remainder: $60 - 4 = 56$ and $76 - 4 = 72$. HCF(56, 72) = 8.

Answer: 8

Example 3. Three bells ring every 12, 15 and 20 minutes. They ring together at 9:00 AM. When will they next ring together?

Solution: LCM(12, 15, 20): $12 = 2^2 \times 3$, $15 = 3 \times 5$, $20 = 2^2 \times 5$. LCM $= 2^2 \times 3 \times 5 = 60$. They ring together again after 60 minutes, at 10:00 AM.

Answer: 10:00 AM

Example 4. Find the smallest number which leaves remainder 5 when divided by 6, 8 and 12.

Solution: LCM(6, 8, 12) = 24. Required number = 24 + 5 = 29.

Answer: 29

Shortcuts

- For two co-prime numbers, HCF = 1 and LCM = their product.
- If the ratio of two numbers is a:b with HCF h, the numbers are ha and hb, and LCM = hab.
- Greatest 4-digit number divisible by x: $9999 - (9999 \bmod x)$. Smallest: $1000 + (x - 1000 \bmod x)$ when 1000 is not divisible.

Common mistakes

- Using HCF when the question wants LCM (check the story).
- Adding the remainder in the HCF pattern instead of subtracting it.
- Applying HCF × LCM = product to three numbers (it works for two only).

Practice questions: LCM and HCF

1. Find the HCF of 24 and 36. Basic
2. Find the LCM of 8, 12 and 18. Basic
3. The HCF of two numbers is 6 and their LCM is 180. One number is 36. Find the other. Basic
4. Find the greatest number that divides 100 and 160 leaving remainders 4 and 10 respectively. Basic
5. Find the smallest number divisible by 15, 20 and 25. Medium

6. Find the HCF of 2/3 and 4/9. Medium

7. Find the smallest 3-digit number divisible by 8, 12 and 15. Medium

8. Find the greatest 4-digit number divisible by 12, 18 and 30. Medium

9. Find the smallest number that leaves remainder 3 when divided by 4, 5 and 6 and is exactly divisible by 7. Advanced

10. Two numbers are in the ratio 3:4 and their LCM is 48. Find their HCF. Advanced

Answers: see the answer key at the end of this guide.

03 NUMERICAL ABILITY · MUST-DO
Percentages

Percent means 'per hundred'. Percentages are the language of profit, discount, interest, data tables and population questions, so speed here pays off across the whole paper.

Start from zero

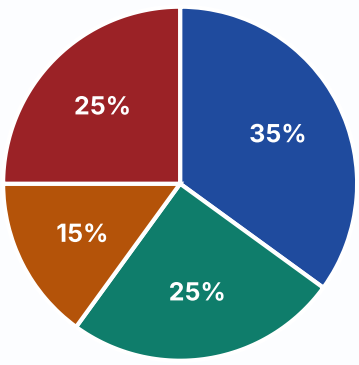
The basic idea

x% means x/100. To find x% of a number N, compute (x/100) × N. To find what percent a is of b, compute (a/b) × 100. Example: 35% of 240 = 0.35 × 240 = 84. And 72 is (72/90) × 100 = 80% of 90.

Fraction to percentage table

Memorise these. They let you do most percentage questions mentally.

Fraction	%	Fraction	%
1/2	50%	1/8	12.5%
1/3	33.33%	1/9	11.11%
1/4	25%	1/10	10%
1/5	20%	1/11	9.09%
1/6	16.67%	1/12	8.33%
1/7	14.28%	1/15	6.67%
1/16	6.25%	1/20	5%



Rent ₹14,000 (35%)
Food ₹10,000 (25%)
Travel ₹6,000 (15%)
Savings ₹10,000 (25%)

Monthly budget ₹40,000

35% of 40,000 = 14,000

Rent : total = 35 : 100 = 7 : 20

A pie chart shows each part as a percentage of 100

Increase and decrease

New value = Old × (1 ± r/100). An increase of 20% means multiply by 1.2. A decrease of 20% means multiply by 0.8. A 20% increase followed by a 20% decrease does **not** return to the start: 1.2 × 0.8 = 0.96, a 4% net fall.

Successive percentage change

For two changes of a% and b%, net change = a + b + (ab/100). Use a plus sign for an increase and a minus sign for a decrease. Example: +20% then −10%: 20 − 10 − 2 = +8%.

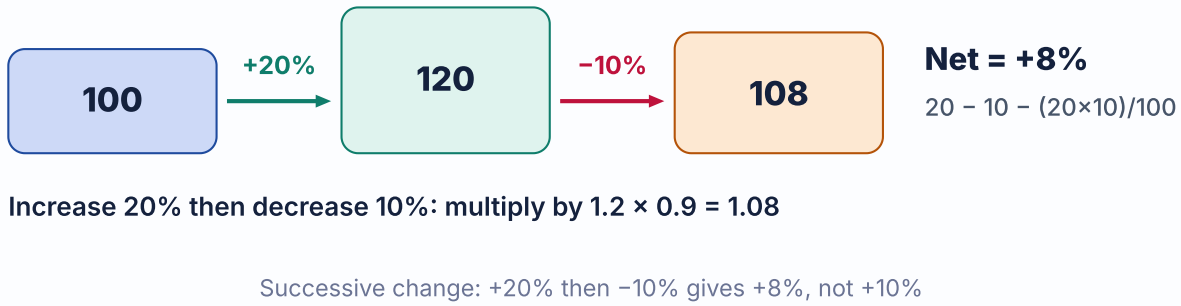
The reverse percentage

If A is x% more than B, then B is [x/(100+x)] × 100 % less than A. If A is x% less than B, then B is [x/(100−x)] × 100 % more than A. Example: A earns 25% more than B, so B earns 25/125 = 20% less than

A.

Price and consumption

If the price rises by r%, a household must cut consumption by [r/(100+r)] × 100 % to keep spending the same. A 25% rise needs a 20% cut. A 20% rise needs a 16.67% cut.



Increase 20% then decrease 10%: multiply by 1.2 × 0.9 = 1.08

Successive change: +20% then −10% gives +8%, not +10%

Formula box

- x% of N = (x/100) × N
- Percentage change = (change / original) × 100
- Successive change = a + b + ab/100
- If A is x% more than B: B is x/(100+x) × 100 % less than A
- Population after n years = $P(1 + r/100)^n$
- Depreciated value = $P(1 - r/100)^n$
- Reduce consumption by r/(100+r) × 100 % when price rises r%

Step-by-step method

1. Convert the percentage to a fraction where possible (25% = 1/4).
2. Identify the base (the 'of' number) before computing.
3. For two or more changes, use the successive-change formula.
4. Check whether the answer should be bigger or smaller than the start.

Solved examples

Example 1. The price of an item rises by 25%. By what percent must a family cut consumption to keep the expenditure the same?

Solution: Cut = 25/(100+25) × 100 = 25/125 × 100 = 20%.

Answer: 20%

Example 2. A number is increased by 20% and then decreased by 10%. Find the net change.

Solution: Net = 20 − 10 + (20 × −10)/100 = 10 − 2 = +8%.

Answer: 8% increase

Example 3. The population of a town is 40,000 and grows 10% per year. Find the population after 2 years.

Solution: $40{,}000 \times (1.1)^2 = 40{,}000 \times 1.21 = 48{,}400$.

Answer: 48,400

Example 4. A candidate gets 45% of votes in a two-candidate election and loses by 300 votes. Find the total votes.

Solution: The winner gets 55%. Difference = 55 − 45 = 10% of total = 300. Total = 3,000.

Answer: 3,000

Shortcuts

- x% of y = y% of x. So 8% of 50 = 50% of 8 = 4.
- To find 15% of a number: 10% + 5% (half of 10%).
- For 'k% more', multiply by (100+k)/100. For 'k% less', multiply by (100−k)/100.

Common mistakes

- Taking the percentage on the wrong base (old vs new value).
- Adding successive percentages directly (20% then −20% is not 0).
- Confusing percentage points with percent (from 40% to 50% is +10 points but +25%).

Practice questions: Percentages

1. Find 15% of 480. Basic
2. 72 is what percent of 90? Basic
3. A value goes from 250 to 300. Find the percentage increase. Basic
4. A number is increased by 20% and then decreased by 20%. Find the net change. Basic
5. If 40% of x is 120, find x. Medium
6. The price of sugar rises by 20%. By what percent should consumption fall to keep expenditure unchanged? Medium
7. A candidate gets 45% of votes and loses by 300 votes (two candidates). Find the total votes. Medium
8. A town of 50,000 people loses 10% of its population every year. What is the population after 2 years? Medium
9. A machine loses 20% of its value each year. After 2 years it is worth ₹64,000. Find its original value. Advanced
10. A is 25% more than B and C is 20% less than A. By what percent is C less or more than B? Advanced

Answers: see the answer key at the end of this guide.

04 NUMERICAL ABILITY · MUST-DO
Profit and Loss

Cost price, selling price, marked price, discount: the vocabulary of every shop. This topic is a favourite because it combines percentages with simple logic.

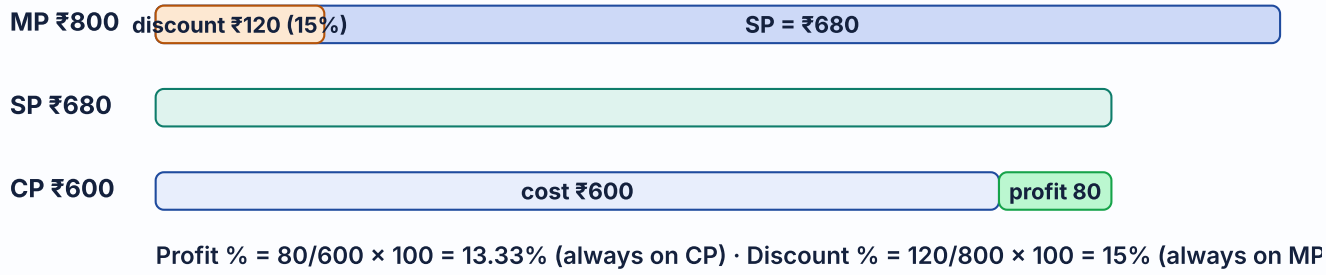
Start from zero

The four key words

Cost Price (CP) is what the seller pays. **Selling Price (SP)** is what the seller receives. **Marked Price (MP)** is the price printed on the tag. **Discount** = MP − SP. If SP > CP there is a profit; if SP < CP there is a loss.

Profit and loss percentages

Profit = SP − CP, and Profit % = (Profit / CP) × 100. Loss % = (Loss / CP) × 100. Always calculate profit or loss percentage on the **cost price**, unless the question says otherwise. Discount percentage is always on the **marked price**.



Marked price, discount, selling price, cost and profit on one scale

Finding SP or CP quickly

SP = CP × (100 + P)/100 for a profit of P%. SP = CP × (100 − L)/100 for a loss of L%. So CP = SP × 100/(100 + P). Example: SP = ₹540 at a 10% loss gives CP = 540 × 100/90 = ₹600.

Discounts

SP = MP × (100 − d)/100. For successive discounts of a% and b%, the single equivalent discount = a + b − ab/100. Example: 20% and 10% make 20 + 10 − 2 = 28%.

False weights

A dishonest dealer sells at cost price but gives less goods. Gain % = [error / (true weight − error)] × 100. Selling 800 g as 1 kg: error 200, gain = 200/800 × 100 = 25%.

Same selling price, equal gain and loss

If two articles are sold at the same price, one at x% gain and the other at x% loss, there is always an overall **loss** of $x^2/100$ %. At 20%: loss = 400/100 = 4%.

Formula box

- Profit % = (SP − CP)/CP × 100
- SP = CP × (100 ± P)/100
- CP = SP × 100/(100 ± P)
- Discount % = (MP − SP)/MP × 100
- Successive discount = a + b − ab/100
- Loss % when equal gain and loss % = $x^2/100$
- Gain % on false weight = error/(true − error) × 100

Step-by-step method

1. List CP, SP, MP, discount clearly.
2. Pick the base for each percentage (CP for profit, MP for discount).
3. Use multiplication factors (1.2, 0.9) to chain steps.
4. If CP is unknown, assume CP = 100 and scale up.

Solved examples

Example 1. An article costing ₹500 is sold for ₹575. Find the profit percentage.

Solution: Profit = 575 − 500 = 75. Profit % = 75/500 × 100 = 15%.

Answer: 15%

Example 2. The marked price is ₹800 and a 15% discount is given. If CP is ₹600, find the profit %.

Solution: SP = 800 × 0.85 = ₹680. Profit = 680 − 600 = 80. Profit % = 80/600 × 100 = 13.33%.

Answer: 13.33%

Example 3. Two successive discounts of 20% and 10% are offered on ₹2,500. Find the final price.

Solution: Net discount = 20 + 10 − 2 = 28%. Price = 2,500 × 0.72 = ₹1,800.

Answer: ₹1,800

Example 4. A shopkeeper marks goods 25% above cost and then gives a 20% discount. Find the profit or loss %.

Solution: Take CP = 100. MP = 125. SP = 125 × 0.8 = 100. Profit = 0.

Answer: No profit, no loss

Shortcuts

- If CP of x articles = SP of y articles, profit % = (x − y)/y × 100.
- Marking up m% and giving d% discount: net % = m − d − md/100.
- Use CP = 100 as an assumed value when only percentages are given.

Common mistakes

- Calculating profit % on the selling price.
- Applying discount on the cost price instead of the marked price.
- Adding successive discounts directly (20% + 10% is not 30%).

Practice questions: Profit and Loss

1. CP = ₹250, SP = ₹300. Find the profit %. Basic
2. An article sold at 10% loss gives SP ₹540. Find the CP. Basic
3. MP = ₹1,500 and discount = 20%. Find the SP. Basic
4. MP = ₹2,000 with successive discounts of 10% and 5%. Find the SP. Basic
5. An article sold at 20% profit. If it had been sold for ₹60 less, profit would have been 10%. Find the CP. Medium
6. Two articles are sold at ₹1,200 each, one at 20% gain and the other at 20% loss. Find the overall result. Medium
7. A dealer uses a 800 g weight for 1 kg but sells at cost price. Find the gain %. Medium
8. CP = ₹400. The shopkeeper marks the item 40% above cost and gives 10% discount. Find the profit %. Medium
9. The CP of 20 articles equals the SP of 16 articles. Find the profit %. Advanced
10. A trader sells an item at 25% profit. If he buys at 20% less and sells ₹10 less, he still makes 25% profit. Find the original CP. Advanced

Answers: see the answer key at the end of this guide.

05 NUMERICAL ABILITY · MUST-DO
Time, Speed and Distance

Speed questions test how well you handle ratios and units. Trains, boats and races are the three standard settings.

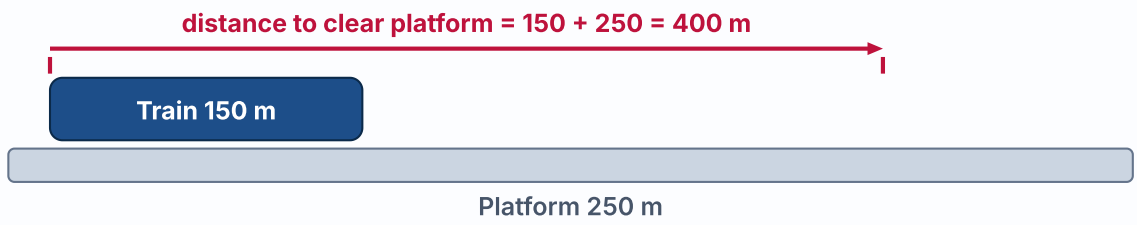
Start from zero

The core relation

Speed = Distance / Time. So Distance = Speed × Time and Time = Distance / Speed. Keep units consistent. To convert km/h to m/s multiply by 5/18. To convert m/s to km/h multiply by 18/5. Example: 72 km/h = 72 × 5/18 = 20 m/s.

Average speed

Average speed = Total distance / Total time. It is **not** the average of the two speeds. If equal distances are covered at speeds x and y, the average speed = 2xy/(x + y). Example: going at 40 and returning at 60 gives 2×40×60/100 = 48 km/h.



Time = 400 ÷ 15 m/s = 26.67 s (54 km/h = 15 m/s)

A train must travel its own length plus the platform

Relative speed

When two objects move in the **same** direction the relative speed is the **difference** of speeds. When they move in **opposite** directions, add the speeds. Relative speed turns a two-body problem into a one-body problem.

Trains

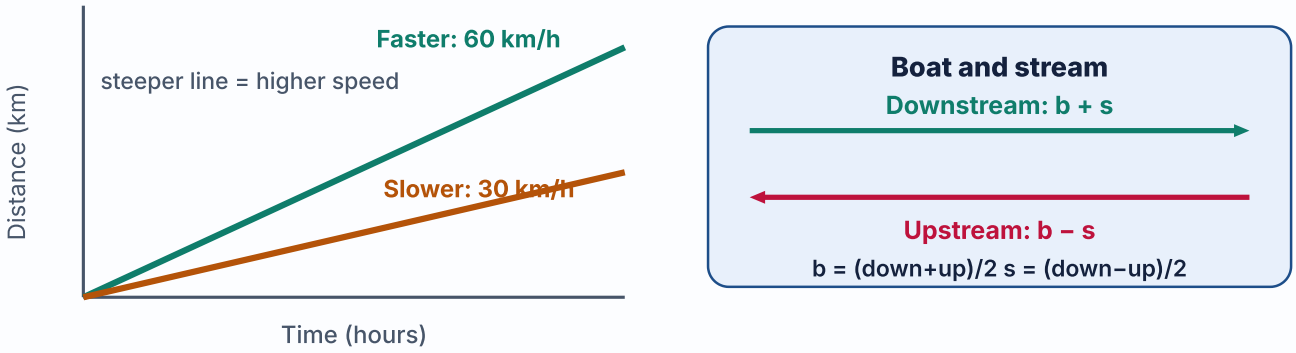
A train crossing a pole or a person travels its own length. A train crossing a platform or bridge travels its length plus the platform length. Two trains crossing each other travel the sum of their lengths at the relative speed.

Boats and streams

Downstream speed = b + s. Upstream speed = b − s. Here b is boat speed in still water and s is the stream speed. So b = (down + up)/2 and s = (down − up)/2.

Speed, time and distance ratios

For a fixed distance, speed and time are inversely proportional. If speed is 4/5 of usual, time is 5/4 of usual. For a fixed time, distance is proportional to speed.



Distance-time graph and boat/stream directions

Formula box

- Speed = Distance/Time
- km/h → m/s: × 5/18; m/s → km/h: × 18/5
- Average speed (equal distances) = 2xy/(x+y)
- Same direction: relative speed = |u − v|; opposite: u + v
- Train crossing a platform: (L + P)/speed
- Two trains: $(L_1 + L_2)$/relative speed
- Boat: b = (d+u)/2, s = (d−u)/2

Step-by-step method

1. Convert all units to a common system before calculating.
2. For trains, decide what distance is covered (own length, or length plus platform).
3. For meeting problems, use the sum of speeds; for chase problems, the difference.
4. Use ratio shortcuts: speed ratio a:b means time ratio b:a for the same distance.

Solved examples

Example 1. A train 150 m long at 54 km/h crosses a platform 250 m long. Find the time taken.

Solution: Speed = 54 × 5/18 = 15 m/s. Distance = 150 + 250 = 400 m. Time = 400/15 = 26.67 s.

Answer: 26.67 s

Example 2. A boat goes 45 km downstream and 45 km upstream. Boat speed in still water is 12 km/h and stream speed is 3 km/h. Find the total time.

Solution: Down = 15 km/h, time = 3 h. Up = 9 km/h, time = 5 h. Total = 8 h.

Answer: 8 hours

Example 3. Two trains of lengths 100 m and 120 m run in opposite directions at 40 km/h and 50 km/h. Find the time to cross each other.

Solution: Relative speed = 90 km/h = 25 m/s. Distance = 220 m. Time = 220/25 = 8.8 s.

Answer: 8.8 s

Example 4. A person walks at 4/5 of his usual speed and reaches 10 minutes late. Find his usual time.

Solution: New time = 5/4 of usual. Extra time = 1/4 of usual = 10 min, so usual time = 40 min.

Answer: 40 minutes

Shortcuts

- 72 km/h = 20 m/s and 54 km/h = 15 m/s. Remember 36 km/h = 10 m/s.
- If a train passes two objects of lengths, subtract the times to get the length difference.
- In a race of 100 m, 'A beats B by 10 m' means B has run 90 m when A finishes.

Common mistakes

- Averaging speeds for the whole journey.
- Forgetting to add the platform length for the train.
- Mixing minutes and hours or km and m.

Practice questions: Time, Speed and Distance

1. Convert 90 km/h into m/s. Basic
2. A car covers 180 km in 3 hours. Find its speed. Basic
3. A 200 m train crosses a pole in 10 s. Find its speed in km/h. Basic
4. A 120 m train at 36 km/h crosses a 180 m bridge. Find the time. Basic
5. Equal distances are covered at 20 km/h and 30 km/h. Find the average speed. Medium
6. A boat goes 30 km downstream in 2 hours and 30 km upstream in 3 hours. Find the stream speed. Medium
7. A starts at 5 km/h, 2 hours before B who starts at 7 km/h from the same place and direction. After how long does B catch A? Medium
8. Two trains of 150 m each run in the same direction at 60 and 45 km/h. Find the time for the faster to cross the slower. Medium
9. A man rows at 6 km/h in still water. It takes him twice as long to go upstream as downstream. Find the stream speed. Advanced
10. A train 300 m long passes a platform in 36 s and a pole in 18 s. Find the platform length. Advanced

Answers: see the answer key at the end of this guide.

06 NUMERICAL ABILITY · MUST-DO

Time and Work

Work problems are rate problems in disguise. Each worker has a rate (work per day), and rates simply add up when people work together.

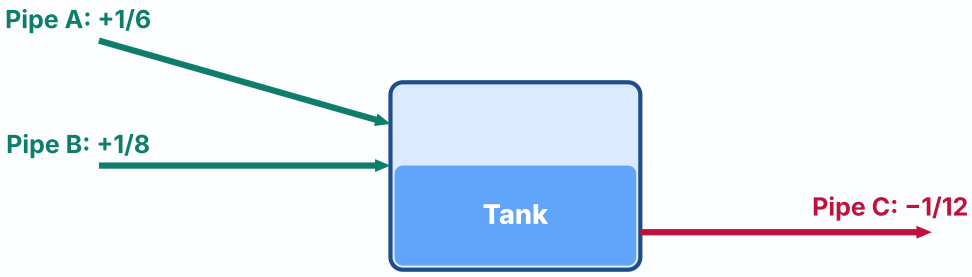
Start from zero

Work rate

If A finishes a job in a days, A's one-day work is 1/a. Together A and B in one day do 1/a + 1/b, so together they need ab/(a + b) days. Example: A in 10 days and B in 15 days: 150/25 = 6 days.

The efficiency (LCM) method

Take the LCM of the days as the total work in 'units'. Then each person's efficiency is units per day. Example: A in 12 days, B in 18, C in 36: total = 36 units. A = 3, B = 2, C = 1 units/day. Together 6 units/day, so 36/6 = 6 days.



Net = 1/6 + 1/8 − 1/12 = 5/24 per hour → 4.8 hours

Pipes add (inlet) or subtract (outlet) their rates

Men, days and hours

Work is proportional to men × days × hours. $M_1D_1H_1 / W_1 = M_2D_2H_2 / W_2$. More men means fewer days (inverse), more work means more days (direct).

Pipes and cisterns

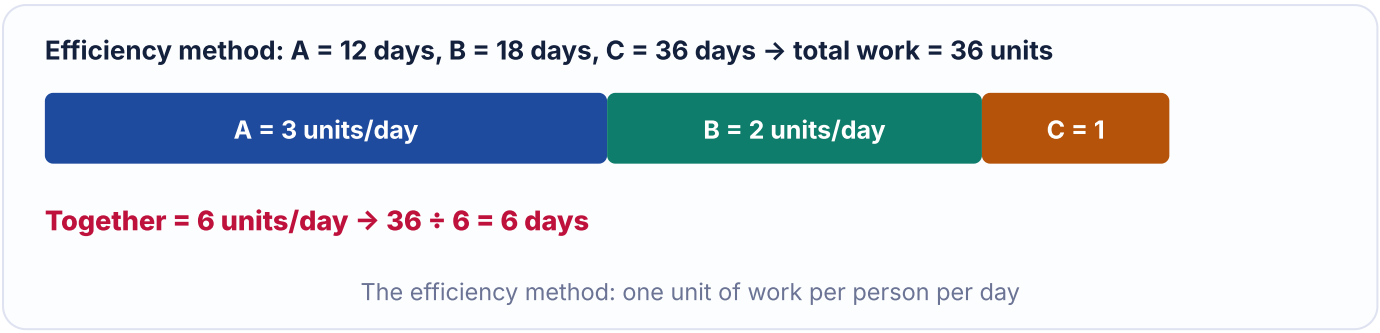
An inlet pipe adds water (positive rate). An outlet or leak removes water (negative rate). Net rate = sum of rates. Example: A fills in 6 h, B in 8 h, C empties in 12 h. Net = 1/6 + 1/8 − 1/12 = 5/24 per hour, so 4.8 h.

Alternate days and leaving early

If A and B work on alternate days, compute one 2-day cycle, count full cycles, then handle the remainder. If someone leaves early, split the total time into stages and add the work done in each stage.

Wages

Money is divided in proportion to the work done, which is rate × days.



The efficiency method: one unit of work per person per day

Formula box

- A and B together = ab/(a+b) days
- If A alone takes a and A+B take t days, B alone takes at/(a − t)
- Work = Men × Days × Hours (proportional)
- Inlet pipe = + rate, outlet pipe = − rate
- Wages ratio = ratio of (rate × days worked)
- Total work = LCM of days (efficiency method)

Step-by-step method

1. Choose total work = LCM of the days.
2. Convert every worker's time into units per day.
3. Add or subtract rates (outlets subtract).
4. Divide the total units by the combined rate.

Solved examples

Example 1. A can finish a job in 10 days and B in 15 days. Find the time taken working together.

Solution: Rates: 1/10 + 1/15 = 5/30 = 1/6 per day. Time = 6 days.

Answer: 6 days

Example 2. A and B together finish a job in 12 days. A alone takes 20 days. How long will B alone take?

Solution: B's rate = 1/12 − 1/20 = (5 − 3)/60 = 1/30. B alone takes 30 days.

Answer: 30 days

Example 3. 8 men build a wall in 12 days. How many days will 6 men need?

Solution: 8 × 12 = 6 × D, so D = 16 days.

Answer: 16 days

Example 4. Pipe A fills a tank in 6 h, pipe B in 8 h and pipe C empties it in 12 h. All three are opened. Find the time to fill it.

Solution: Net rate = 1/6 + 1/8 − 1/12 = (4 + 3 − 2)/24 = 5/24 per hour. Time = 24/5 = 4.8 h.

Answer: 4.8 hours

Shortcuts

- Work done in a part: time/total time gives the fraction (3/5 of work takes 3/5 of the time).
- For 'A and B alternate', one cycle = rate of A + rate of B.
- If efficiency ratio is a:b, time ratio is b:a.

Common mistakes

- Adding days instead of adding daily work rates.
- Forgetting to subtract an outlet pipe.
- Splitting wages in the ratio of days instead of work done.

Practice questions: Time and Work

1. A can do a job in 12 days and B in 24 days. Find the time working together. **Basic**
2. A does a job in 15 days. How many days will 3/5 of the job take? **Basic**
3. 10 men finish a job in 15 days. How many men will finish it in 10 days? **Basic**
4. A and B together take 12 days; A alone takes 20 days. How long does B alone take? **Basic**
5. A pipe fills a tank in 5 h and a leak empties a full tank in 15 h. Find the time to fill the tank with the leak. **Medium**
6. A takes 20 days and B takes 30 days. They work on alternate days starting with A. Find the total days. **Medium**
7. A+B finish in 10 days, B+C in 15 days, A+C in 12 days. Find the days for A+B+C together. **Medium**
8. A and B together are paid ₹900 for a job. A alone can do it in 10 days and B alone in 15 days. Divide the wages fairly. **Medium**
9. 5 men or 10 women finish a job in 24 days. How many days will 3 men and 4 women take? **Advanced**
10. A and B together finish in 6 days and A alone in 10 days. Find the days B alone needs. **Advanced**

Answers: see the answer key at the end of this guide.

07 NUMERICAL ABILITY · MUST-DO

Ratio and Proportion

A ratio compares two quantities. A proportion says two ratios are equal. These ideas support partnership, mixtures, ages and almost every word problem.

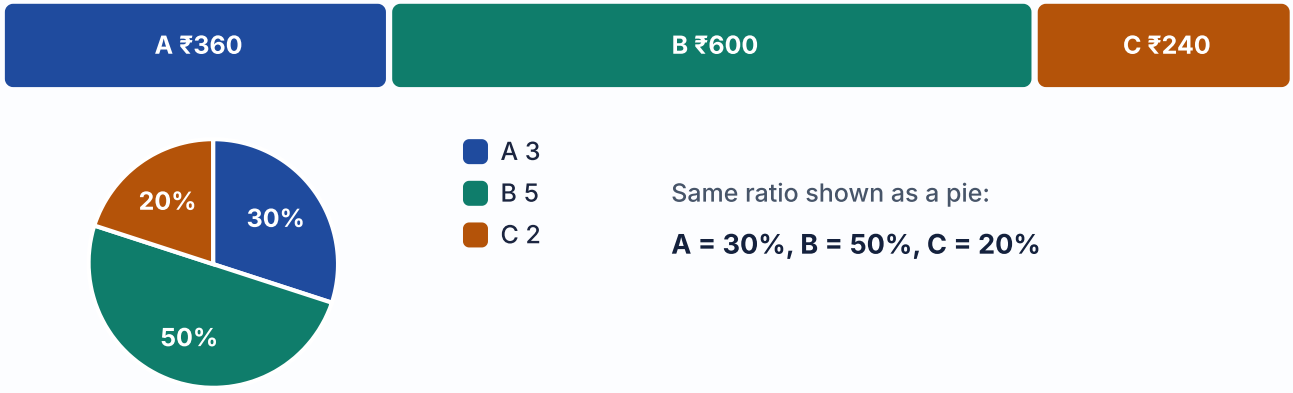
Start from zero

Ratio

The ratio of a to b is written a : b = a/b. Both quantities must be in the same unit. Simplify by dividing both sides by their HCF. Example: 24 : 36 = 2 : 3.

Combining ratios

If A : B = 2 : 3 and B : C = 4 : 5, make B equal in both: A : B = 8 : 12, B : C = 12 : 15. So A : B : C = 8 : 12 : 15. The compound ratio of a:b and c:d is ac : bd.



₹1,200 split 3 : 5 : 2 as a bar and as a pie

Dividing a quantity

To split an amount N in the ratio a : b : c, find total parts (a + b + c). Each part = N/(a + b + c). Then multiply. Example: ₹1,200 in 3 : 5 : 2 gives ₹360, ₹600 and ₹240.

Proportion

If a : b = c : d then ad = bc. Here d is the fourth proportional. The mean proportional between a and b is √(ab). The third proportional to a and b is b^2/a.

Direct and inverse proportion

In direct variation, if one quantity doubles, the other doubles (cost vs quantity). In inverse variation, if one doubles, the other halves (speed vs time, men vs days).

Useful algebra

If a/b = c/d then (a + b)/(a − b) = (c + d)/(c − d) (componendo and dividendo). If ratio is 3 : 4, assume actual values 3x and 4x and solve with the given total or difference.

Formula box

- a : b = a/b
- Compound ratio of a:b and c:d = ac : bd
- If a:b = c:d then ad = bc
- Mean proportional = √(ab)
- Third proportional to a, b = b^2/a
- Amount to each = N × (part / total parts)
- Direct: $a_1/b_1 = a_2/b_2$; Inverse: $a_1b_1 = a_2b_2$

Step-by-step method

1. Express unknowns as multiples (3x, 4x).
2. Use the given sum or difference to find x.
3. Make common terms equal when combining ratios.
4. For new ratios after adding a value, form an equation.

Solved examples

Example 1. Divide ₹1,200 among A, B and C in the ratio 3 : 5 : 2.

Solution: Total parts = 10. One part = 120. A = 360, B = 600, C = 240.

Answer: ₹360, ₹600, ₹240

Example 2. If A : B = 2 : 3 and B : C = 4 : 5, find A : B : C.

Solution: Make B the same: A : B = 8 : 12 and B : C = 12 : 15. So A : B : C = 8 : 12 : 15.

Answer: 8 : 12 : 15

Example 3. The ratio of boys to girls is 7 : 5 and the total is 360. If 30 more girls join, find the new ratio.

Solution: Boys = 210, girls = 150. After 30 girls join: 210 : 180 = 7 : 6.

Answer: 7 : 6

Example 4. Two numbers are in the ratio 5 : 7. If 6 is added to each, the ratio becomes 8 : 11. Find the numbers.

Solution: (5x + 6)/(7x + 6) = 8/11. 55x + 66 = 56x + 48, so x = 18. Numbers are 90 and 126.

Answer: 90 and 126

Shortcuts

- If x : y = a : b, the value of x/(x+y) is a/(a+b).
- For three-term ratios with a common middle term, multiply cross.
- If 2A = 3B = 4C, then A : B : C = 1/2 : 1/3 : 1/4 = 6 : 4 : 3.

Common mistakes

- Comparing quantities in different units.
- Not making the common term equal before combining ratios.
- Treating 'ratio 3 : 4' as actual values 3 and 4.

Practice questions: Ratio and Proportion

1. Divide ₹900 in the ratio 2 : 3 : 4. Basic
2. If A : B = 3 : 4 and B : C = 2 : 5, find A : B : C. Basic
3. Find x if 3 : 5 = x : 20. Basic
4. Find the mean proportional between 4 and 16. Basic
5. Two numbers are in the ratio 3 : 4 and their sum is 84. Find the numbers. Medium
6. A gets ₹160, which is 4 parts in the ratio 4 : 5 with B. Find the total amount. Medium
7. Income : expenditure = 5 : 4 and savings are ₹4,000. Find the income. Medium
8. 6 pens cost ₹90. Find the cost of 10 pens. Medium
9. Two numbers are in the ratio 5 : 7. After adding 6 to each the ratio is 8 : 11. Find the numbers. Advanced
10. If 2A = 3B = 4C, find A : B : C. Advanced

Answers: see the answer key at the end of this guide.

08 NUMERICAL ABILITY · IMPORTANT
Averages

An average describes a whole group with a single number. Average questions are fast marks if you remember to work with totals rather than averages.

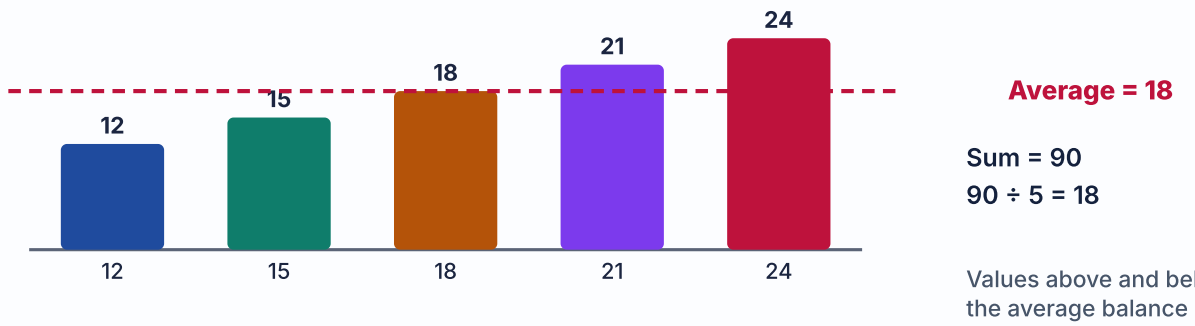
Start from zero

The definition

Average = Sum of all values / Number of values. So Sum = Average × Number. Always convert averages into sums; sums are easy to add and subtract.

Averages of series

The average of the first n natural numbers is (n+1)/2. For the first n odd numbers it is n. For the first n even numbers it is (n+1). For consecutive numbers, the average is the middle value.



The average (dashed line) balances the values above and below it

Weighted average

When groups have different sizes, weight each average by its size. Weighted average = $(n_1a_1 + n_2a_2 + \ldots)/(n_1 + n_2 + \ldots)$. Example: 20 students at average 60 and 30 at average 70 gives (1,200 + 2,100)/50 = 66.

Adding, removing, replacing

Adding a new value: new sum = old sum + new value. Removing: subtract. Replacing a value by another changes the sum by (new − old); the average changes by (new − old)/n.

Average speed and ages

Average speed uses total distance/total time (see Time, Speed and Distance). For ages, when everyone gets older by t years, the average also rises by t years. When a person joins or leaves, recompute with the total age.

Formula box

- Average = Sum/Number of items
- Sum = Average × Number
- Average of first n natural numbers = (n+1)/2
- Weighted average = $\Sigma(n_ia_i)/\Sigma n_i$
- Change in average on replacing = (new − old)/n
- New value joining = old average + n × (change in average)

Step-by-step method

1. Convert each average into a sum.
2. Add or subtract sums according to the story.
3. Divide by the new count to get the new average.
4. Use change-in-average shortcuts for replacements.

Solved examples

Example 1. The average of 5 numbers is 20. A sixth number is added and the average becomes 22. Find the sixth number.

Solution: Old sum = 100. New sum = 22 × 6 = 132. Sixth number = 132 − 100 = 32.

Answer: 32

Example 2. Class A has 20 students with average 60. Class B has 30 students with average 70. Find the combined average.

Solution: Sum = 20×60 + 30×70 = 1,200 + 2,100 = 3,300. Average = 3,300/50 = 66.

Answer: 66

Example 3. The average age of 10 persons is 30. A 45-year-old person is replaced by a 25-year-old. Find the new average.

Solution: Change in sum = 25 − 45 = −20. New average = 30 − 20/10 = 28.

Answer: 28

Example 4. A batsman's average after 10 innings is 40. He scores 62 in the 11th innings. Find his new average.

Solution: Total = 400 + 62 = 462. New average = 462/11 = 42.

Answer: 42

Shortcuts

- The weighted average always lies between the two group averages, closer to the larger group.
- Average of an AP = (first + last)/2.
- If every value increases by k, the average increases by k.

Common mistakes

- Averaging the averages without weights.
- Forgetting to update the count when a value is added or removed.
- Dividing by the old number of items.

Practice questions: Averages

1. Find the average of the first 10 natural numbers. Basic
2. The average of four numbers is 15. Three of them are 10, 12 and 18. Find the fourth. Basic
3. The average of 6 numbers is 25. The average of the first four is 20. Find the average of the last two. Basic
4. A batsman has an average of 40 in 10 innings. He scores 62 in his next innings. Find the new average. Basic

5. The average of 5 consecutive numbers is 20. Find the smallest. Medium

6. The average of 5 numbers is 18. When one is removed the average is 16. Find the removed number. Medium

7. The average weight of 8 people increases by 2 kg when a person of 65 kg is replaced by a new one. Find the weight of the new person. Medium

8. Find the average of the first five odd numbers. Medium

9. 25 kg of rice at ₹40/kg is mixed with 15 kg at ₹60/kg. Find the average price per kg. Advanced

10. The average age of 5 family members is 24 years and the youngest is 4 years old. Find the average age of the other four. Advanced

Answers: see the answer key at the end of this guide.

09

NUMERICAL ABILITY · IMPORTANT

Simple and Compound Interest

Interest is the price of using money. Simple interest grows in a straight line; compound interest grows faster because interest earns interest.

Start from zero

Simple Interest

SI = P × R × T / 100, where P is the principal, R is the yearly rate in percent and T is the time in years. Amount A = P + SI. Example: ₹5,000 at 8% for 3 years gives SI = 5,000 × 8 × 3/100 = ₹1,200.

Compound Interest

Interest is added to the principal every period. $A = P(1 + R/100)^n$ and CI = A − P. Example: ₹10,000 at 10% for 2 years: A = 10,000 × 1.21 = ₹12,100, so CI = ₹2,100 (SI would be ₹2,000).

₹10,000 at 10% per year: SI grows in a straight line, CI curves upward



SI grows as a line; CI curves upward because interest earns interest

Different compounding periods

For half-yearly compounding use rate R/2 and 2n periods. For quarterly use R/4 and 4n. Example: ₹8,000 at 10% for 1 year compounded half-yearly: 5% for 2 periods = 8,000 × 1.1025 = ₹8,820.

CI and SI difference

For 2 years the difference CI − SI = $P(R/100)^2$. For 3 years it is $P(R/100)^2 \times (300 + R)/100$. Example: ₹8,000 at 5% for 2 years: 8,000 × $(0.05)^2$ = ₹20.

Doubling and tripling

With SI, a sum becomes n times itself in 100(n−1)/R years. At 20% SI a sum doubles in 5 years. Under compound interest, use the rule of 72: years to double ≈ 72/R.

Formula box

- SI = PRT/100
- A = P + SI
- $A = P(1 + R/100)^n$ (compound)
- CI = A − P
- Half-yearly: rate R/2, periods 2n
- Quarterly: rate R/4, periods 4n
- $CI - SI\ (2\ yrs) = P(R/100)^2$
- $CI - SI\ (3\ yrs) = P(R/100)^2(300 + R)/100$

Step-by-step method

1. Check whether interest is simple or compound.
2. Match the rate period with the compounding period.
3. Use the multiplier (1 + R/100) for each period.
4. For differences, use the shortcut formulas.

Solved examples

Example 1. Find the SI on ₹5,000 at 8% per year for 3 years.

Solution: SI = 5,000 × 8 × 3/100 = ₹1,200. Amount = ₹6,200.

Answer: SI ₹1,200

Example 2. Find the compound interest on ₹10,000 at 10% for 2 years.

Solution: $A = 10{,}000 \times (1.1)^2 = 12{,}100$. CI = 12,100 − 10,000 = ₹2,100.

Answer: ₹2,100

Example 3. A sum doubles in 5 years at simple interest. Find the rate.

Solution: SI = P in 5 years: R = 100/5 = 20% per year.

Answer: 20%

Example 4. ₹8,000 is invested at 10% per year, compounded half-yearly, for 1 year. Find the amount.

Solution: Rate per period 5%, periods 2. $A = 8{,}000 \times (1.05)^2 = 8{,}000 \times 1.1025 = ₹8{,}820$.

Answer: ₹8,820

Shortcuts

- The multipliers 10% → 1.1, 20% → 1.2, 5% → 1.05 make compounding quick.
- For 2 years at r% CI, the net growth is $2r + r^2/100$ percent.
- For CI differences use the shortcut, not the full calculation.

Common mistakes

- Using R directly when compounding half-yearly (use R/2 and 2n).
- Calculating CI as the final amount without subtracting the principal.
- Mixing months and years in T.

Practice questions: Simple and Compound Interest

1. Find the SI on ₹2,000 at 5% for 4 years. **Basic**
2. Find the principal if SI is ₹360 at 6% for 3 years. **Basic**
3. Find the compound interest on ₹5,000 at 20% for 2 years. **Basic**
4. Find the difference between CI and SI on ₹8,000 at 5% for 2 years. **Basic**
5. At what rate will ₹4,000 become ₹4,840 in 2 years at CI? **Medium**
6. In how many years does a sum double at 12.5% simple interest? **Medium**
7. A sum becomes ₹6,050 in 2 years at 10% CI. Find the sum. **Medium**
8. Find the amount on ₹10,000 at 20% for 1 year compounded half-yearly. **Medium**
9. ₹3,000 is split into two parts lent at 4% and 6% SI for 1 year, yielding total interest ₹150. Find the part lent at 4%. **Advanced**
10. Find CI − SI on ₹10,000 at 10% for 3 years. **Advanced**

Answers: see the answer key at the end of this guide.

10 NUMERICAL ABILITY · IMPORTANT
Probability

Probability measures how likely an event is. At TCS NQT level you need the sample space of coins, dice and cards, and two simple rules: add for 'or' (mutually exclusive) and multiply for 'and' (independent).

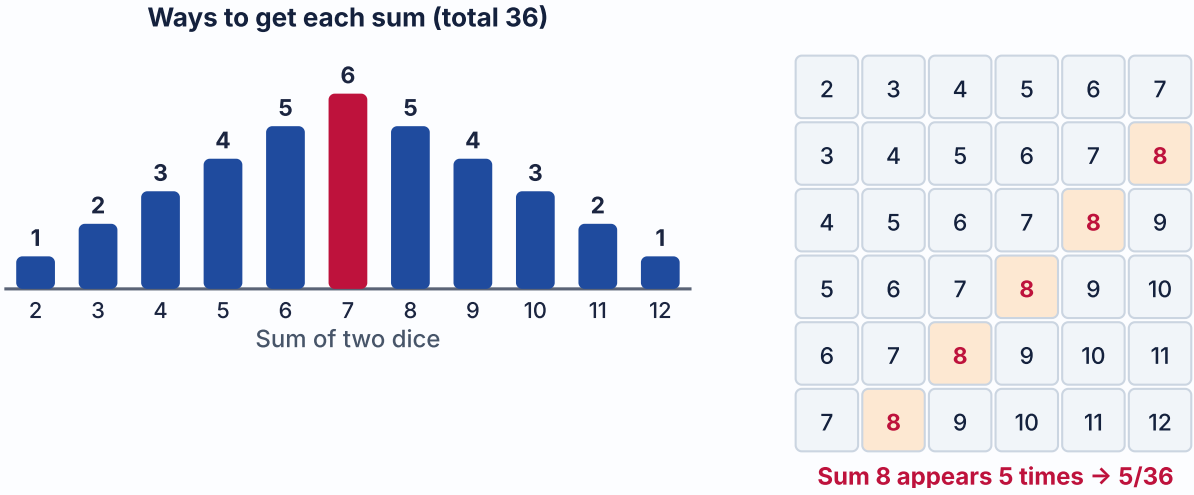
Start from zero

Basic definition

P(E) = favourable outcomes / total outcomes. Probability lies between 0 (impossible) and 1 (certain). The probability that an event does not happen is 1 − P(E).

Coins

A coin has 2 outcomes. n coins have 2^n outcomes. For 3 coins the outcomes are HHH, HHT, HTH, HTT, THH, THT, TTH, TTT. P(exactly two heads) = 3/8.



Two dice: ways to get each sum, and the 6 × 6 outcome grid

Dice

One die has 6 outcomes. Two dice have 36. The number of ways to get each sum is shown in the chart below. Sum 7 is the most likely (6 ways, probability 1/6).

Cards

A deck has 52 cards in 4 suits (hearts, diamonds red; spades, clubs black), each with 13 cards (A, 2 ... 10, J, Q, K). There are 12 face cards, 4 aces, 4 kings and 26 red cards.

Rules

OR rule: P(A or B) = P(A) + P(B) − P(A and B). If A and B cannot happen together, the last term is zero.
AND rule: for independent events P(A and B) = P(A) × P(B). For drawing without replacement, the second probability changes: P(both red) = (5/8) × (4/7) for 5 red out of 8.

At least one

Use the complement: P(at least one) = 1 − P(none). Example: at least one head in 3 tosses = 1 − 1/8 = 7/8.

A deck: 4 suits × 13 cards = 52 (26 red, 26 black, 12 face cards, 4 aces)

♥	A	2	3	4	5	6	7	8	9	10	J	Q	K
♦	A	2	3	4	5	6	7	8	9	10	J	Q	K
♠	A	2	3	4	5	6	7	8	9	10	J	Q	K
♣	A	2	3	4	5	6	7	8	9	10	J	Q	K

Red 26

Black 26

P(king or heart) = 4/52 + 13/52 − 1/52 = 16/52 = 4/13

(king of hearts is counted in both groups, so subtract it once)

A standard deck of 52 cards

Formula box

- P(E) = favourable/total
- P(not E) = 1 − P(E)
- P(A or B) = P(A) + P(B) − P(A and B)
- P(A and B) = P(A) × P(B) (independent)
- n coins → 2^n outcomes; 2 dice → 36 outcomes
- P(at least one) = 1 − P(none)
- Cards: 52 total, 26 red, 12 face cards, 4 aces

Step-by-step method

1. List the total outcomes (sample space).
2. Count favourable outcomes carefully.
3. Use the complement for 'at least one'.
4. For multi-stage events, multiply stage probabilities, changing counts if there is no replacement.

Solved examples

Example 1. Two dice are rolled. Find the probability that the sum is 8.

Solution: Favourable: (2,6), (3,5), (4,4), (5,3), (6,2) = 5 outcomes. Probability = 5/36.

Answer: 5/36

Example 2. A card is drawn from a deck. Find the probability that it is a king or a heart.

Solution: Kings 4, hearts 13, king of hearts counted twice. Favourable = 4 + 13 − 1 = 16. Probability = 16/52 = 4/13.

Answer: 4/13

Example 3. A bag has 5 red and 3 blue balls. Two balls are drawn without replacement. Find P(both red).

Solution: P = (5/8) × (4/7) = 20/56 = 5/14.

Answer: 5/14

Example 4. A coin is tossed 3 times. Find P(at least one head).

Solution: P(no head) = $(1/2)^3$ = 1/8. P(at least one head) = 1 − 1/8 = 7/8.

Answer: 7/8

Shortcuts

- Memorise the dice-sum counts: 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 for sums 2 to 12.
- P(at least one) = 1 − P(none) is almost always faster.
- With replacement means the probabilities stay the same.

Common mistakes

- Counting (3,5) and (5,3) as one outcome for dice.
- Forgetting to subtract the overlap in the OR rule.
- Not reducing the fraction at the end (TITA answers need the exact form requested).

Practice questions: Probability

1. Find the probability of getting an even number on a die. **Basic**
2. Two dice are thrown. Find P(sum = 7). **Basic**
3. Find the probability of drawing an ace from a deck. **Basic**
4. Two coins are tossed. Find P(both tails). **Basic**
5. A bag has 4 red and 6 green balls. Find P(green). **Medium**
6. Find the probability of a prime number on a die. **Medium**
7. Three coins are tossed. Find P(exactly 2 heads). **Medium**
8. Two cards are drawn without replacement. Find P(both are aces). **Medium**
9. Two dice are rolled. Find P(at least one six). **Advanced**
10. Two dice are rolled. Find P(sum ≥ 10). **Advanced**

Answers: see the answer key at the end of this guide.

11 NUMERICAL ABILITY · IMPORTANT
Permutation and Combination

P and C is the mathematics of counting. It answers questions like 'in how many ways can we arrange or choose?'. The one decision to master is: does order matter (permutation) or not (combination)?

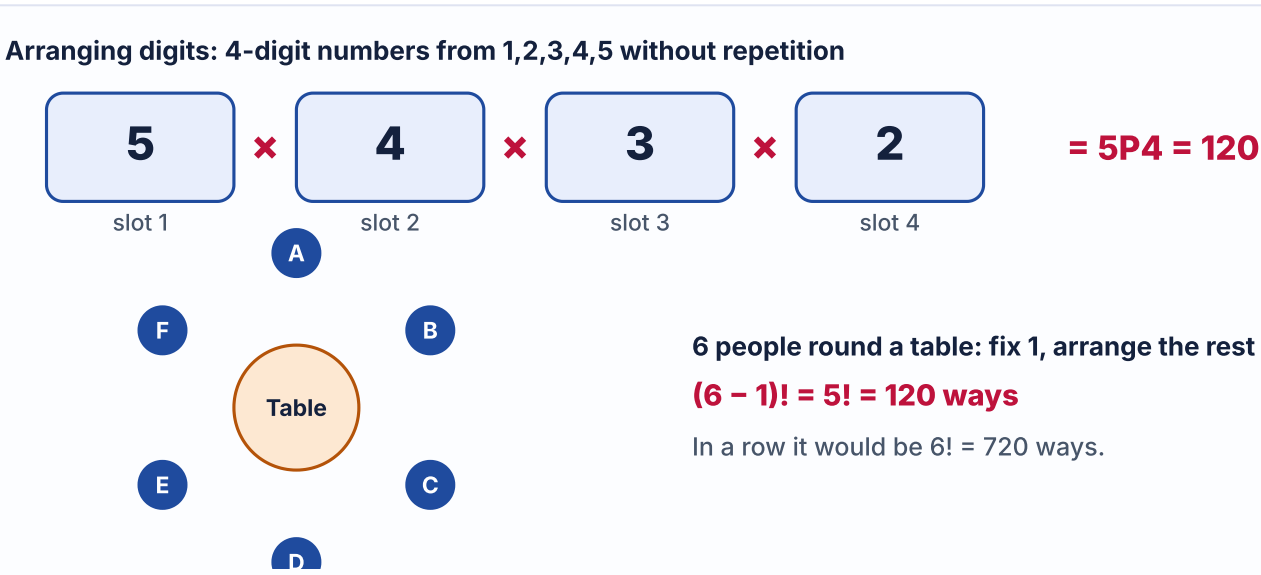
Start from zero

Counting principle

If one task can be done in m ways and a second in n ways, then doing both in order can be done in m × n ways. For three tasks it is m × n × p. Example: 3 shirts and 4 trousers give 12 outfits.

Factorial

n! = n × (n−1) × ... × 1, and 0! = 1. So 5! = 120. The number of ways to arrange n different objects in a row is n!.



Slot method for arrangements, and circular seating

Permutation (order matters)

nPr = n!/(n−r)!. It counts arrangements of r objects out of n. Example: 8P3 = 8 × 7 × 6 = 336.

Combination (order does not matter)

nCr = n!/[r!(n−r)!]. It counts selections. Note nCr = nC(n−r). Example: choosing a committee of 3 from 8: 8C3 = 56.

Repeated letters and circles

For n objects where p are alike and q are alike, arrangements = n!/(p! q!). 'LEVEL' has 5 letters with L twice and E twice: 5!/(2!2!) = 30. For n people around a circular table, arrangements = (n−1)!.

Useful applications

Handshakes among n people = nC2. Diagonals of an n-sided polygon = nC2 − n. Number of ways to select with a restriction: fix the condition first, then choose from the rest.

Formula box

- n! = n(n−1)(n−2)...1
- nPr = n!/(n−r)!
- nCr = n!/[r!(n−r)!]
- nCr = nC(n−r)
- Arrangements with repetition = n!/(p!q!...)
- Circular arrangements = (n−1)!
- Handshakes = nC2; Diagonals = nC2 − n

Step-by-step method

1. Decide: arranging (order matters) or selecting (order does not)?
2. Handle restrictions first (fix a person, glue a pair).
3. Use nCr for groups, nPr or n! for lines.
4. For repeated letters divide by factorials of the repeats.

Solved examples

Example 1. In how many ways can 5 different books be arranged on a shelf?

Solution: 5! = 120.

Answer: 120

Example 2. A committee of 2 men and 2 women is chosen from 5 men and 4 women. Find the number of ways.

Solution: 5C2 × 4C2 = 10 × 6 = 60.

Answer: 60

Example 3. How many ways can the letters of LEVEL be arranged?

Solution: 5 letters with L twice and E twice: 5!/(2! × 2!) = 120/4 = 30.

Answer: 30

Example 4. In how many ways can 6 people sit around a circular table?

Solution: (6 − 1)! = 120.

Answer: 120

Shortcuts

- nC2 = n(n−1)/2 and nC3 = n(n−1)(n−2)/6.
- Always compute nCr using the smaller of r and n−r.
- 'At least' problems: total ways minus ways with none.

Common mistakes

- Using permutation when only selection is needed.
- Forgetting to divide for repeated letters.
- Using n! instead of (n−1)! for circular seating.

Practice questions: Permutation and Combination

1. Find 6!. Basic
2. Find 10C2. Basic
3. Find 7P2. Basic
4. How many arrangements does the word APPLE have? Basic
5. How many 4-digit numbers can be formed from 1, 2, 3, 4, 5 without repetition? Medium
6. 12 people shake hands with each other once. How many handshakes? Medium
7. A committee of 2 men and 2 women is formed from 5 men and 4 women. How many ways? Medium
8. In how many ways can 5 persons sit around a round table? Medium
9. How many arrangements does MISSISSIPPI have? Advanced
10. In how many ways can 3 people be selected from 6 if a particular person must be included? Advanced

Answers: see the answer key at the end of this guide.

12 Mensuration

NUMERICAL ABILITY · IMPORTANT

Mensuration is the measurement of lengths, areas and volumes. Memorise the formulas once, keep units consistent (area in square units, volume in cubic units) and use π = 22/7 unless told otherwise.

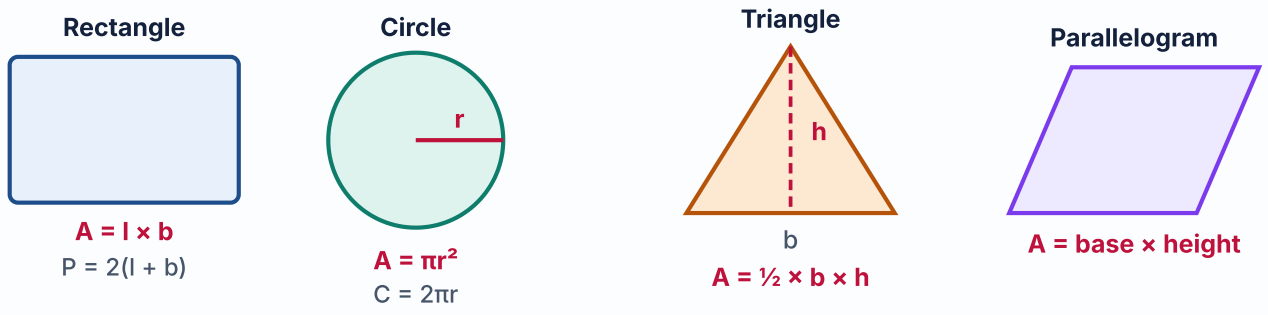
Start from zero

2D shapes: perimeter and area

Shape	Area	Perimeter
Rectangle (l, b)	l × b	2(l + b)
Square (a)	a^2	4a
Triangle	½ × base × height	sum of sides
Circle (r)	πr^2	$2\pi r$
Parallelogram	base × height	2(a + b)
Trapezium	½(a + b) × h	sum of sides
Rhombus	½ × d_1 × d_2	4a

3D shapes: surface area and volume

Solid	Curved/Total surface	Volume
Cube (a)	TSA = $6a^2$	a^3
Cuboid (l, b, h)	TSA = 2(lb + bh + hl)	l × b × h
Cylinder (r, h)	CSA = $2\pi rh$; TSA = $2\pi r(r + h)$	$\pi r^2 h$
Cone (r, h, slant l)	CSA = πrl; TSA = $\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Sphere (r)	$4\pi r^2$	$(4/3)\pi r^3$
Hemisphere	CSA = $2\pi r^2$; TSA = $3\pi r^2$	$(2/3)\pi r^3$



2D shapes

Slant height and diagonals

For a cone, slant height l = $\sqrt{(r^2 + h^2)}$. A right triangle with sides 3, 4, 5 appears often (r = 3, h = 4 gives l = 5). The diagonal of a cuboid = $\sqrt{(l^2 + b^2 + h^2)}$. The diagonal of a cube = $a\sqrt{3}$.

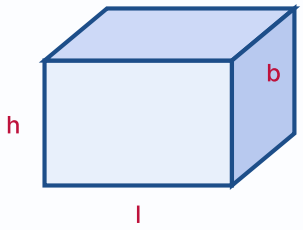
Recasting and melting

When a solid is melted and reshaped, the **volume** stays the same. A sphere of radius 6 melted into small spheres of radius 2 gives $(6/2)^3 = 27$ small spheres.

Practical problems

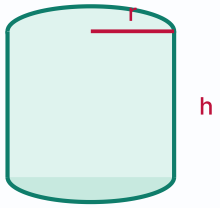
Fencing and boundary cost use perimeter. Painting, flooring and tiling use area. Water tanks, filling and pouring use volume. Convert units first: 1 m = 100 cm; 1 m^3 = 1000 litres.

Cuboid V = lbh



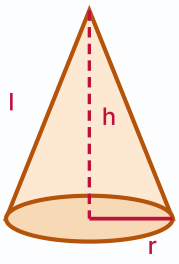
TSA = 2(lb+bh+hl)

Cylinder V = πr²h



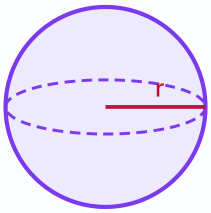
CSA = 2πrh

Cone V = ⅓πr²h



l = √(r² + h²), CSA = πrl

Sphere V = (4/3)πr³



SA = 4πr²

Hemisphere TSA = 3πr²

3D solids with the radius, height and slant height marked

Formula box

- Circle: A = πr², C = 2πr
- Triangle: A = ½bh; Heron: √[s(s−a)(s−b)(s−c)]
- Cylinder: V = πr²h, CSA = 2πrh
- Cone: V = ⅓πr²h, l = √(r²+h²), CSA = πrl
- Sphere: V = (4/3)πr³, SA = 4πr²
- Cuboid: V = lbh; diagonal = √(l²+b²+h²)
- Cube: V = a³, TSA = 6a²

Step-by-step method

1. Draw a quick sketch and label measurements.
2. Pick the formula (area, surface or volume).
3. Use π = 22/7 when radius is a multiple of 7.
4. Check units at the end.

Solved examples

Example 1. Find the area and circumference of a circle of radius 7 cm.

Solution: Area = 22/7 × 49 = 154 cm^2. Circumference = 2 × 22/7 × 7 = 44 cm.

Answer: 154 cm^2, 44 cm

Example 2. Find the volume of a cylinder with radius 7 cm and height 10 cm.

Solution: V = 22/7 × 49 × 10 = 1,540 cm^3.

Answer: 1,540 cm^3

Example 3. A cone has radius 3 cm and height 4 cm. Find its slant height and volume (in terms of π).

Solution: $l = \sqrt{(9 + 16)} = 5$ cm. $V = \frac{1}{3} \times \pi \times 9 \times 4 = 12\pi$ cm³ ≈ 37.7 cm³.

Answer: l = 5 cm, V = 12π

Example 4. A metallic sphere of radius 6 cm is melted into smaller spheres of radius 2 cm. How many are made?

Solution: Volume ratio = $(6/2)^3 = 27$.

Answer: 27

Shortcuts

- Doubling the radius multiplies the circle's area by 4 and the sphere's volume by 8.
- Cost problems: Area or perimeter × rate.
- A rectangle with fixed perimeter has maximum area when it is a square.

Common mistakes

- Using diameter instead of radius.
- Mixing square and cubic units.
- Using height instead of slant height for the curved surface area of a cone.

Practice questions: Mensuration

1. A triangle has base 12 cm and height 9 cm. Find its area. **Basic**
2. A circle has diameter 14 cm. Find its area. **Basic**
3. A cube has side 5 cm. Find its total surface area. **Basic**
4. A cuboid measures 5 × 4 × 3 cm. Find its volume. **Basic**
5. A trapezium has parallel sides 10 cm and 14 cm and height 6 cm. Find its area. **Medium**
6. A cylinder has radius 3.5 cm and height 10 cm. Find its curved surface area. **Medium**
7. Find the volume of a sphere of radius 21 cm. **Medium**
8. A rhombus has diagonals 16 cm and 12 cm. Find its area. **Medium**
9. Find the cost of fencing a circular field of radius 14 m at ₹10 per metre. **Advanced**
10. Find the volume of a cone with radius 7 cm and height 6 cm. **Advanced**

Answers: see the answer key at the end of this guide.

13 NUMERICAL ABILITY · IMPORTANT
Mixtures and Alligation

Alligation is a fast graphical method to mix two ingredients to a required average. It is just the weighted average idea turned into a ratio trick.

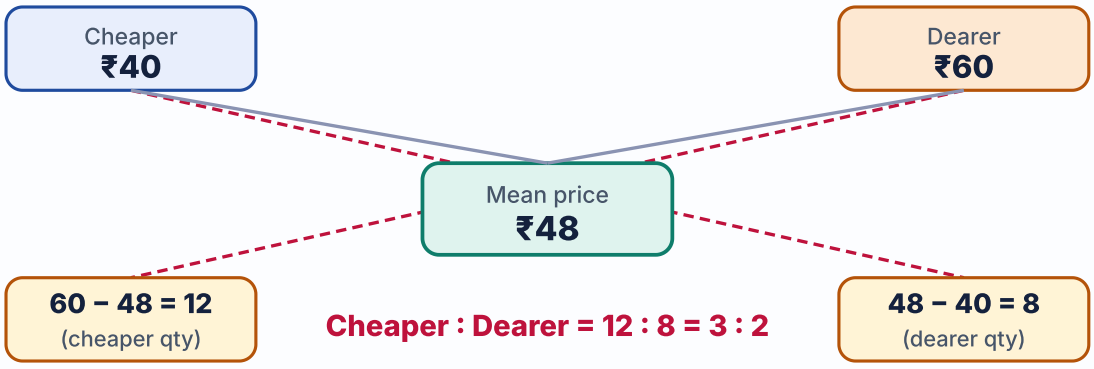
Start from zero

The alligation rule

If you mix a cheaper item (price c) with a dearer item (price d) to get a mixture of mean price m, then **(quantity of cheaper) : (quantity of dearer) = (d − m) : (m − c)**. Draw the cross: put c and d on top, m in the middle, subtract diagonally.

Mixing solutions

The same rule works for percentages of any ingredient (acid in water, milk in water). Treat each concentration like a price. Example: mixing 30% and 50% solutions to get 40% needs equal quantities because 50 − 40 = 40 − 30.



The alligation cross: subtract diagonally to get the mixing ratio

Replacement of a liquid

A vessel has V litres of pure milk. If x litres are removed and replaced with water, and this is done n times, milk left = $V \times (1 - x/V)^n$. Example: 20 L, remove 4 L twice: $20 \times (0.8)^2 = 12.8$ L.

Adding or removing water

The quantity of the other ingredient stays the same while water changes. Use that fixed quantity to find the new ratio. If 50 L of a 20% solution contains 10 L alcohol, adding water until it is 10% means 100 L total, so add 50 L water.

Cost and profit in mixing

Mixing cheap and costly items to sell at a certain price gives a profit or loss. First find the mean cost price using the alligation, then apply the profit % on it.

Formula box

- Cheaper : Dearer = (d − m) : (m − c)
- Milk left after n replacements = $V(1 - x/V)^n$
- Mean value = $(n_1a_1 + n_2a_2)/(n_1 + n_2)$
- Concentration = (quantity of ingredient/total) × 100
- Ratio after adding water: ingredient stays constant

Step-by-step method

1. Identify the cheap and costly values and the target mean.
2. Write the cross diagram and subtract diagonally.
3. Convert the resulting ratio into quantities.
4. For replacement, keep the fraction removed each time.

Solved examples

Example 1. In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture at ₹48/kg?

Solution: Cheaper : Dearer = (60 − 48) : (48 − 40) = 12 : 8 = 3 : 2.

Answer: 3 : 2

Example 2. 20 L of pure milk has 4 L removed and replaced by water. This is done twice. Find the milk left.

Solution: $20 \times (1 - 4/20)^2 = 20 \times 0.64 = 12.8$ L.

Answer: 12.8 L

Example 3. 50 L of a 20% alcohol solution is given. How much water should be added to make it 10%?

Solution: Alcohol = 10 L. For 10%, total = 100 L. Water to add = 100 − 50 = 50 L.

Answer: 50 L

Example 4. 4 L of a 10% solution is mixed with 6 L of a 20% solution. Find the concentration of the mixture.

Solution: Alcohol = 0.4 + 1.2 = 1.6 L in 10 L = 16%.

Answer: 16%

Shortcuts

- Equal quantities give the simple average of the two values.
- The mean always lies between the two prices (closer to the larger quantity).
- For one replacement with equal quantities, milk left = V(1 − x/V).

Common mistakes

- Swapping the order of the cross subtraction.
- Changing the ingredient quantity when adding water.
- Applying the replacement formula with the wrong volume x.

Practice questions: Mixtures and Alligation

1. Two teas at ₹300 and ₹400 per kg are mixed to sell at ₹360. Find the ratio (cheaper : dearer). **Basic**
2. 4 L of 10% and 6 L of 20% solutions are mixed. Find the concentration. **Basic**
3. 36 L of milk and water in the ratio 5 : 1. How much water should be added to make it 5 : 2? **Basic**
4. Sugar at ₹20 and ₹28 per kg is mixed 1 : 1. Find the mean price. **Basic**
5. A vessel has 40 L of milk. 10 L is replaced by water, and this is repeated once more. Find the milk left. **Medium**
6. In what ratio is wheat at ₹12 mixed with wheat at ₹16 to get ₹13.5? **Medium**
7. In what ratio should 30% and 50% acid solutions be mixed to get 40%? **Medium**
8. 60 L of milk and water is in the ratio 2 : 1. 20 L of water is added. Find the new ratio of milk to water. **Medium**
9. 40 kg of a 10% salt solution is evaporated until it is 20% salt. How much water evaporates? **Advanced**
10. A 16 L vessel is full of milk. 4 L is removed and replaced by water, and this is repeated once. Find the milk left. **Advanced**

Answers: see the answer key at the end of this guide.

14 NUMERICAL ABILITY · GOOD TO KNOW
Ages, Simplification and Approximation

Three small topics that give quick marks: age puzzles, BODMAS simplification and estimating answers when options are far apart.

Start from zero

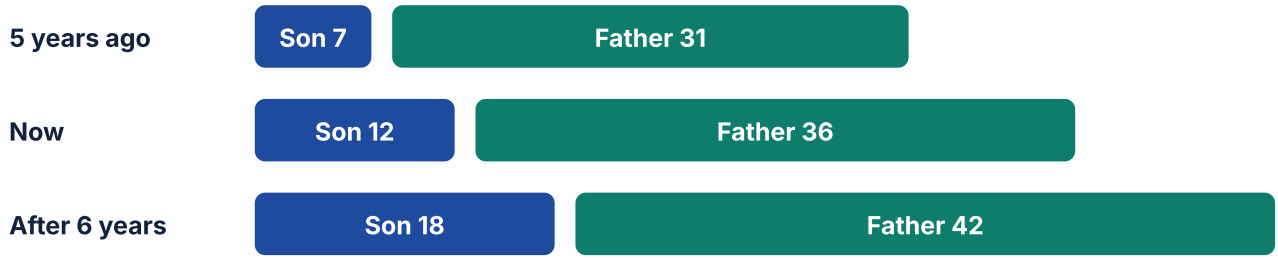
Age problems

Assign variables to present ages and write one equation for each statement. Remember: after t years, add t to every age. t years ago, subtract t. The difference between two persons' ages never changes. Example: father is 3 times son and the sum is 48, so son = 12, father = 36.

Typical age statements

'Five years ago A was twice B' → A − 5 = 2(B − 5). 'After 10 years the ratio becomes 5 : 4' → (A + 10)/(B + 10) = 5/4. 'The sum of ages is 40' → A + B = 40.

Ages move together on a timeline: the difference never changes



Father − Son = 24 years at every moment. Now: 36 = 3 × 12 ✔

Ages move together on a timeline; the age difference stays fixed

BODMAS

Brackets, Orders (powers and roots), Division and Multiplication (left to right), Addition and Subtraction (left to right). Example: $18 \div 3 \times 2 + 5^2 - 4 = 6 \times 2 + 25 - 4 = 33$.

Squares, cubes and roots

Memorise squares to 30 and cubes to 15 (see the cheat sheet). Roots of perfect squares then become instant. Example: $\sqrt{1296} = 36$.

Approximation

When options are far apart, round numbers to friendly values. Example: 48.7% of 623 ≈ 50% of 620 = 310 (exact 303.4), good enough to choose between 250, 303 and 360.

Formula box

- Age after t years = present age + t
- Age difference stays constant
- BODMAS order: Brackets, Orders, D/M, A/S
- $(a + b)^2 = a^2 + 2ab + b^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $\sqrt{(a^2)} = a$ for $a \geq 0$

Step-by-step method

1. Define variables for present ages.
2. Write one equation for each sentence.
3. Solve the equations (substitution or elimination).
4. Check the answer against the original statements.

Solved examples

Example 1. The sum of the ages of a father and son is 48 years and the father is three times as old as the son. Find their ages.

Solution: Let son = x. Father = 3x. 4x = 48, x = 12. Son 12, father 36.

Answer: Son 12, father 36

Example 2. Five years ago A was twice as old as B. Now their ages sum to 40. Find the present ages.

Solution: A − 5 = 2(B − 5) and A + B = 40. So A = 2B − 5. Then 3B − 5 = 40, B = 15, A = 25.

Answer: A = 25, B = 15

Example 3. Simplify $18 \div 3 \times 2 + 5^2 - 4$.

Solution: 18 ÷ 3 = 6, 6 × 2 = 12, $5^2 = 25$. 12 + 25 − 4 = 33.

Answer: 33

Example 4. Estimate 48.7% of 623.

Solution: ≈ 50% of 620 = 310; the exact value is 303.4.

Answer: ≈ 303

Shortcuts

- For ratio age problems, use x as the multiplier.
- In age sum problems, each year later adds (number of people) to the sum.
- For approximation, round to the nearest ten or hundred.

Common mistakes

- Forgetting to age all persons by the same number of years.
- Ignoring left-to-right order for division and multiplication.
- Over-rounding when options are close.

Practice questions: Ages, Simplification and Approximation

1. A father is 30 and his son is 6. After how many years will the father be three times the son's age? Basic
2. The sum of the ages of two persons is 50 and their difference is 10. Find the ages. Basic
3. Two ages are in the ratio 4 : 5 and their sum is 45. Find the ages. Basic
4. Ten years ago a father was 4 times as old as his son. Now the father is twice the son's age. Find the present ages. Basic
5. A is 6 years older than B. After 4 years the sum of their ages will be 38. Find A's present age. Medium

6. Simplify 8 + 12 ÷ 4 × 3 − 5. Medium

7. Find 15% of 80 + 25% of 48. Medium

8. Find $(0.5)^2 + (0.3)^2$. Medium

9. Find $\sqrt{1296}$. Advanced

10. Find 3/4 of 5/6 of 240. Advanced

Answers: see the answer key at the end of this guide.

Formula cheat sheet

Every key formula of the section on a few pages. Revise it daily in the last two weeks.

1. Number System

- Number of factors of $N = p^a q^b r^c \rightarrow (a+1)(b+1)(c+1)$
- Sum of factors = $[(p^{a+1}-1)/(p-1)] \times [(q^{b+1}-1)/(q-1)] \times \ldots$
- Trailing zeros of n! = $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \ldots$
- Dividend = Divisor × Quotient + Remainder
- Sum of first n natural numbers = $n(n+1)/2$
- Sum of first n odd numbers = n^2; first n even numbers = $n(n+1)$
- $(a+b)^2 = a^2+2ab+b^2$, $(a-b)^2 = a^2-2ab+b^2$, $a^2-b^2 = (a+b)(a-b)$

2. LCM and HCF

- HCF × LCM = product of the two numbers
- HCF ≤ every given number ≤ LCM
- HCF of fractions = HCF(numerators) / LCM(denominators)
- LCM of fractions = LCM(numerators) / HCF(denominators)
- Smallest number leaving remainder r = LCM + r
- Greatest number leaving remainder r = HCF(a−r, b−r, ...)

3. Percentages

- x% of N = $(x/100) \times N$
- Percentage change = (change / original) × 100
- Successive change = $a + b + ab/100$
- If A is x% more than B: B is $x/(100+x) \times 100$ % less than A
- Population after n years = $P\,(1 + r/100)^n$
- Depreciated value = $P\,(1 - r/100)^n$
- Reduce consumption by $r/(100+r) \times 100$ % when price rises r%

4. Profit and Loss

- Profit % = $(SP - CP)/CP \times 100$
- $SP = CP \times (100 \pm P)/100$
- $CP = SP \times 100/(100 \pm P)$
- Discount % = $(MP - SP)/MP \times 100$
- Successive discount = $a + b - ab/100$
- Loss % when equal gain and loss % = $x^2/100$
- Gain % on false weight = error/(true − error) × 100

5. Time, Speed and Distance

- Speed = Distance/Time
- km/h → m/s: × 5/18; m/s → km/h: × 18/5
- Average speed (equal distances) = $2xy/(x+y)$
- Same direction: relative speed = $|u - v|$; opposite: $u + v$
- Train crossing a platform: $(L + P)$/speed
- Two trains: $(L_1 + L_2)$/relative speed
- Boat: $b = (d+u)/2$, $s = (d-u)/2$

6. Time and Work

- A and B together = $ab/(a+b)$ days
- If A alone takes a and A+B take t days, B alone takes $at/(a - t)$
- Work = Men × Days × Hours (proportional)
- Inlet pipe = + rate, outlet pipe = − rate
- Wages ratio = ratio of (rate × days worked)
- Total work = LCM of days (efficiency method)

7. Ratio and Proportion

- $a : b = a/b$
- Compound ratio of a:b and c:d = ac : bd
- If a:b = c:d then ad = bc
- Mean proportional = $\sqrt{(ab)}$
- Third proportional to a, b = b^2/a
- Amount to each = N × (part / total parts)
- Direct: $a_1/b_1 = a_2/b_2$; Inverse: $a_1b_1 = a_2b_2$

8. Averages

- Average = Sum/Number of items
- Sum = Average × Number
- Average of first n natural numbers = $(n+1)/2$
- Weighted average = $\Sigma(n_i a_i)/\Sigma n_i$
- Change in average on replacing = (new − old)/n
- New value joining = old average + n × (change in average)

9. Simple and Compound Interest

- SI = PRT/100
- A = P + SI
- $A = P(1 + R/100)^n$ (compound)
- CI = A − P
- Half-yearly: rate R/2, periods 2n
- Quarterly: rate R/4, periods 4n
- CI − SI (2 yrs) = $P(R/100)^2$
- CI − SI (3 yrs) = $P(R/100)^2(300 + R)/100$

10. Probability

- P(E) = favourable/total
- P(not E) = 1 − P(E)
- P(A or B) = P(A) + P(B) − P(A and B)
- P(A and B) = P(A) × P(B) (independent)
- n coins → 2^n outcomes; 2 dice → 36 outcomes
- P(at least one) = 1 − P(none)
- Cards: 52 total, 26 red, 12 face cards, 4 aces

11. Permutation and Combination

- n! = n(n−1)(n−2)...1
- nPr = n!/(n−r)!
- nCr = n!/[r!(n−r)!]
- nCr = nC(n−r)
- Arrangements with repetition = n!/(p!q!...)
- Circular arrangements = (n−1)!
- Handshakes = nC2; Diagonals = nC2 − n

12. Mensuration

- Circle: $A = \pi r^2$, $C = 2\pi r$
- Triangle: A = ½bh; Heron: $\sqrt{[s(s-a)(s-b)(s-c)]}$
- Cylinder: $V = \pi r^2 h$, $CSA = 2\pi rh$
- Cone: $V = \frac{1}{3}\pi r^2 h$, $l = \sqrt{(r^2+h^2)}$, $CSA = \pi rl$
- Sphere: $V = (4/3)\pi r^3$, $SA = 4\pi r^2$
- Cuboid: V = lbh; diagonal = $\sqrt{(l^2+b^2+h^2)}$
- Cube: $V = a^3$, $TSA = 6a^2$

13. Mixtures and Alligation

- Cheaper : Dearer = (d − m) : (m − c)
- Milk left after n replacements = $V(1 - x/V)^n$
- Mean value = $(n_1a_1 + n_2a_2)/(n_1 + n_2)$
- Concentration = (quantity of ingredient/total) × 100
- Ratio after adding water: ingredient stays constant

14. Ages, Simplification and Approximation

- Age after t years = present age + t
- Age difference stays constant
- BODMAS order: Brackets, Orders, D/M, A/S
- $(a + b)^2 = a^2 + 2ab + b^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $\sqrt{(a^2)} = a$ for $a \geq 0$

Speed tables

Squares 1 to 30 and cubes 1 to 15

n	n^2	n^3	n	n^2	n^3
1	1	1	16	256	–
2	4	8	17	289	–
3	9	27	18	324	–
4	16	64	19	361	–
5	25	125	20	400	–
6	36	216	21	441	–
7	49	343	22	484	–
8	64	512	23	529	–
9	81	729	24	576	–
10	100	1000	25	625	–
11	121	1331	26	676	–
12	144	1728	27	729	–
13	169	2197	28	784	–
14	196	2744	29	841	–
15	225	3375	30	900	–

Fractions as percentages

Fraction	%	Fraction	%
1/2	50%	1/9	11.11%
1/3	33.33%	1/10	10%
1/4	25%	1/11	9.09%
1/5	20%	1/12	8.33%
1/6	16.67%	1/13	7.69%
1/7	14.28%	1/14	7.14%
1/8	12.5%	1/15	6.67%
2/3	66.67%	1/16	6.25%
3/4	75%	1/20	5%
3/8	37.5%	1/25	4%
5/8	62.5%	1/40	2.5%
7/8	87.5%	1/50	2%

Primes up to 100

> 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 (25 primes)

Unit conversions

From	To
km/h → m/s	× 5/18
m/s → km/h	× 18/5
1 hour	60 min = 3600 s
1 m^3	1000 litres
1 km	1000 m
1 m	100 cm

Common speeds in m/s

km/h	m/s
18	5
36	10
54	15
72	20
90	25
108	30

Multiplication factors

Change	Multiply by
+5% / −5%	1.05 / 0.95
+10% / −10%	1.10 / 0.90
+20% / −20%	1.20 / 0.80
+25% / −25%	1.25 / 0.75
+50% / −50%	1.50 / 0.50

Mini test: 20 questions in 25 minutes

Take this without looking at the guide. Target: 15 or more correct. No negative marking.

1. The unit digit of 13^{47} is?
2. Find the HCF of 84 and 126.
3. Find 18% of 450.
4. An article bought for ₹800 is sold for ₹920. Find the profit %.
5. A 90 m train at 36 km/h passes a pole. Find the time in seconds.
6. A can do a job in 18 days and B in 9 days. Together, how many days?
7. Two numbers are in the ratio 2 : 3 and their sum is 75. Find the larger number.
8. The average of 5 numbers is 30. One number, 50, is removed. Find the new average.
9. Find the simple interest on ₹6,000 at 7% per year for 2 years.
10. Two dice are rolled. Find P(sum = 9).
11. Find 7C3.
12. Find the volume of a cylinder of radius 7 cm and height 5 cm.
13. 3 L of a 20% solution is mixed with 2 L of a 45% solution. Find the concentration.
14. A mother is 4 times as old as her daughter. The sum of their ages is 50. Find the daughter's age.
15. MP = ₹1,000, discount 10%, CP = ₹800. Find the profit %.
16. Find the remainder when 2^{20} is divided by 3.
17. A boat's speed is 8 km/h in still water and the stream is 2 km/h. Find the time to go 30 km upstream.
18. The price rises by 50%. By what % should consumption fall to keep expenditure same?
19. Find the LCM of 12, 16 and 24.
20. In how many ways can 6 people sit in a row if two particular people must sit together?

Answer key

1. Number System

1. 3
2. 2
3. Yes (alternating sum 22)
4. 18
5. 12
6. 101101
7. 1008
8. 1
9. 01
10. 56

2. LCM and HCF

1. 12
2. 72
3. 30
4. 6
5. 300
6. 2/9
7. 120
8. 9900
9. 63
10. 4

3. Percentages

1. 72
2. 80%
3. 20%
4. 4% decrease
5. 300
6. 16.67% (1/6)
7. 3,000
8. 40,500
9. ₹1,00,000
10. 0% (equal)

4. Profit and Loss

1. 20%
2. ₹600
3. ₹1,200
4. ₹1,710
5. ₹600
6. 4% loss
7. 25%
8. 26%
9. 25%
10. ₹40

5. Time, Speed and Distance

1. 25 m/s
2. 60 km/h
3. 72 km/h
4. 30 s
5. 24 km/h
6. 2.5 km/h
7. 5 hours after B starts
8. 72 s
9. 2 km/h
10. 300 m

6. Time and Work

1. 8 days
2. 9 days
3. 15 men
4. 30 days
5. 7.5 hours
6. 24 days
7. 8 days
8. A ₹540, B ₹360
9. 24 days
10. 15 days

7. Ratio and Proportion

1. ₹200, ₹300, ₹400
2. 3 : 4 : 10
3. 12
4. 8

5. 36 and 48
6. ₹360
7. ₹20,000
8. ₹150
9. 90 and 126
10. 6 : 4 : 3

8. Averages

1. 5.5
2. 20
3. 35
4. 42
5. 18
6. 26
7. 81 kg
8. 5
9. ₹47.5
10. 29 years

9. Simple and Compound Interest

1. ₹400
2. ₹2,000
3. ₹2,200
4. ₹20
5. 10%
6. 8 years
7. ₹5,000
8. ₹12,100
9. ₹1,500
10. ₹310

10. Probability

1. 1/2
2. 1/6
3. 1/13
4. 1/4
5. 3/5
6. 1/2
7. 3/8
8. 1/221
9. 11/36
10. 1/6

11. Permutation and Combination

1. 720
2. 45
3. 42
4. 60
5. 120
6. 66
7. 60
8. 24
9. 34,650
10. 10

12. Mensuration

1. 54 cm^2
2. 154 cm^2
3. 150 cm^2
4. 60 cm^3
5. 72 cm^2
6. 220 cm^2
7. 38,808 cm^3
8. 96 cm^2
9. ₹880
10. 308 cm^3

13. Mixtures and Alligation

1. 2 : 3
2. 16%
3. 6 L
4. ₹24
5. 22.5 L
6. 5 : 3
7. 1 : 1
8. 1 : 1
9. 20 kg
10. 9 L

14. Ages, Simplification and Approximation

1. 6 years
2. 30 and 20

3. 20 and 25
4. Son 15, father 30
5. 18
6. 12
7. 24
8. 0.34
9. 36
10. 150

Mini test answers

1. 7
2. 42
3. 81
4. 15%
5. 9 s
6. 6 days
7. 45
8. 25
9. ₹840
10. 1/9
11. 35
12. 770 cm^3
13. 30%
14. 10 years
15. 12.5%
16. 1
17. 5 hours
18. 33.33%
19. 48
20. 240